

Anomaly Detection with Conditional VAEs

Borealis AI

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Overview

- 1 Introduction
- 2 Variational Autoencoders
- 3 Conditional VAEs
- 4 Conclusions and Future Work

Plan

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- 2 Variational Autoencoders
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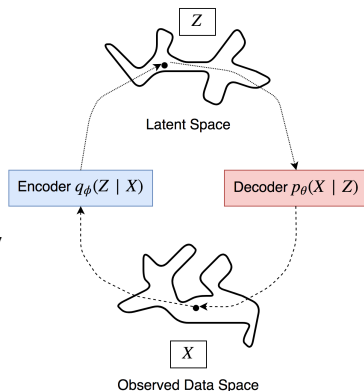
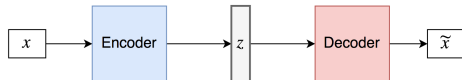
- Anomalies/Outliers - data points not following the general trend
- **Goals**
 - Detection of novel articles from an incoming news stream
 - Relevant to Apollo
 - Filtering anomalous news - value adding to financial analysts
- Variational Autoencoders
 - Generative models with interesting properties

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Autoencoding (Deterministic)

- Obtain a compressed representation of the data x from which it is possible to re-construct it
- Encoder $q_\phi(z|x)$ and Decoder $p_\theta(x|z)$ are jointly trained to maximize the conditional log-likelihood
- The latent representation z has an arbitrary distribution

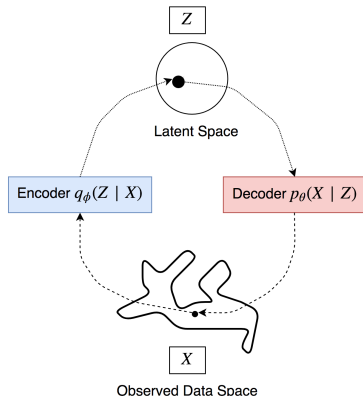
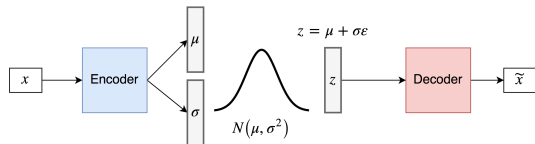


Minimize Reconstruction Loss

$$J = - \sum_{n=1}^N \log p(x^{(n)}|z^{(n)})$$

Variational Autoencoder [Kingma and Welling, 2013]

- Enforce a distribution on the latent space
- Minimize the Kullback-Leibler (KL) divergence between the learnt posterior and a pre-specified prior: $\text{KL}(\mathcal{N}(\mu, \sigma) || \mathcal{N}(0, I))$
- Balance between reconstruction and KL penalty term
 - High λ - Ignores reconstruction
 - Low λ - Deterministic behaviour



Minimize Reconstruction Loss + KL Divergence

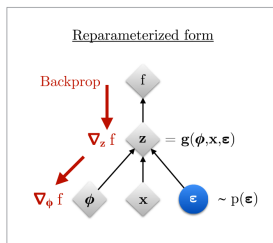
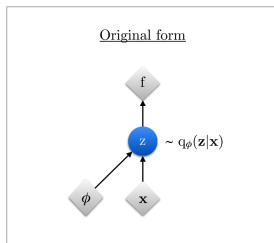
$$J = \sum_{n=1}^N \left[- \mathbb{E}_{z^{(n)} \sim q} [\log p(x^{(n)} | z^{(n)})] + \lambda \cdot \text{KL}(q(z^{(n)} | x^{(n)}) || p(z)) \right]$$

Reparameterization Trick

KL Divergence between posterior and standard normal prior

$$\text{KL}(\mathcal{N}(\mu, \sigma) || \mathcal{N}(0, I)) = \frac{1}{2}(1 + \log((\sigma^{(n)})^2) - (\mu^{(n)})^2 - (\sigma^{(n)})^2)$$

- Model training via SGD and error backpropagation
- Cannot sample directly from the approximate posterior distribution $\mathcal{N}(\mu, \sigma)$
- Stochastic Node - disconnect in the graph
- **Solution:** Sample from fixed distribution $\mathcal{N}(0, I)$ and reparameterize
- $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \otimes \epsilon$ where $\epsilon \sim \mathcal{N}(0, I)$



MNIST Experiments

- Toy Example - Compress image to 2d latent space and reconstruct

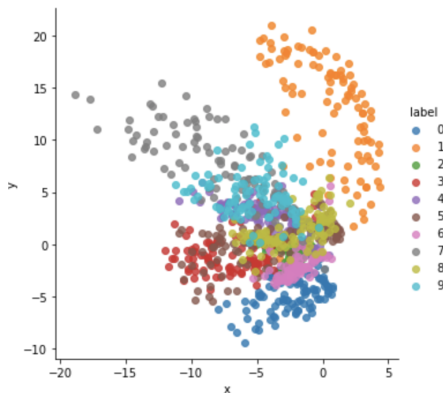


Figure: Deterministic AE

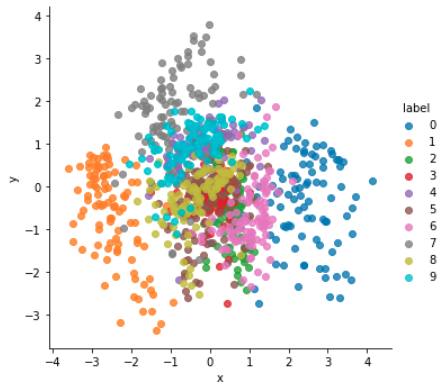


Figure: Variational AE

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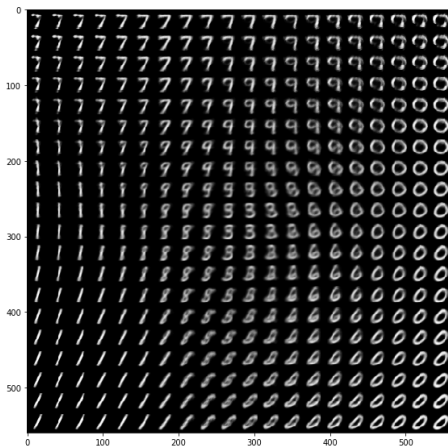


Figure: VAE Reconstructions from different parts of the latent space

Text VAEs

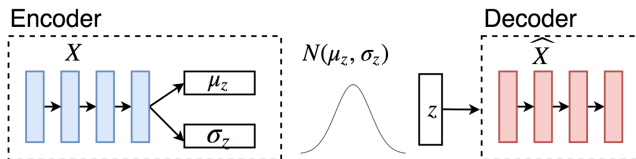


Figure: Model Architecture

- Trained on a subset of SNLI Dataset [Bowman et al., 2015a]

Training Heuristics

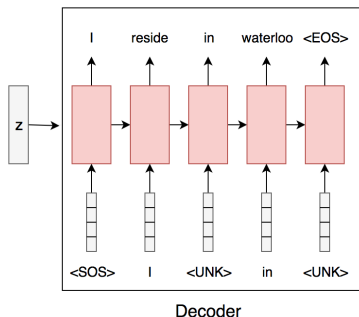
- Training VAEs for text generation is notoriously difficult
- Adopt two training strategies [Bowman et al., 2015b]

KL Weight Annealing

- Gradually increase λ from zero to a threshold value
- Deterministic autoencoder $\rightarrow$ Variational autoencoder
- Experiment with different annealing schedules

Word Dropout

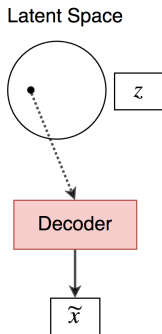
- Replace decoder inputs with $\langle \text{UNK} \rangle$ with probability p
- Weakens the decoder and encourages the model to encode more information into z



Random Sampling

- VAEs exhibit interesting properties due to their learnt latent space
- Continuous latent space $\implies$ meaningful sentences
- Discard encoder; Sample from prior $\mathcal{N}(0, I)$ and generate
- New and interesting sentences unseen in the training data

Deterministic AE	Variational AE
<i>a men wears an umbrella waits to a couple cows a monument there is sleeping and two rug . a man in a pick photos a boy are people at a lake escape .</i>	<i>the dog is sleeping in the grass . the girls are being detained . the group of people are going to begin . a girl with blond-hair on a bike with a stick a woman and a man are walking on a street</i>

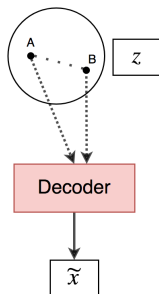


Linear Interpolation

- To test the continuity of the latent space
- $\mathbf{z}_{\alpha_i} = \alpha_i \cdot \mathbf{z}_A + (1 - \alpha_i) \cdot \mathbf{z}_B$ where $\alpha_i \in [0, \frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5}, 1]$
- VAE - Smooth transition maintaining syntax and semantics
- DAE - Transition is irregular and non-continuous

Deterministic AE	Variational AE
Sentence A: there is a couple eating cake .	
<i>there is a couple eating cake .</i> <i>there is a couple eating cake .</i> <i>there is a couple eating cake .</i> <i>there is a group of people eating a party .</i> <i>a group of men are watching a party .</i> <i>a group of men are watching a dance party .</i> <i>a group of men are watching a dance party .</i> <i>a group of men are watching a dance party .</i>	<i>there is a couple eating cake .</i> <i>there is a couple eating .</i> <i>there is a couple eating dinner .</i> <i>there is a couple of people eating dinner .</i> <i>a group of people are having a conversation .</i> <i>a group of men are having a discussion .</i> <i>a group of men are watching a movie .</i> <i>a group of men are watching a movie theater .</i>
Sentence B: a group of men are watching a dance party .	

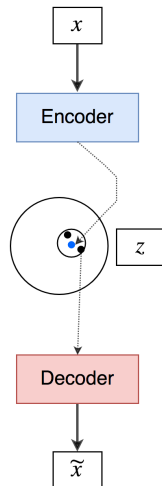
Latent Space



Sampling from Neighborhood

- For a given input x , sample the latent vector as $z = \mu + 3\sigma \otimes \epsilon$
- VAE - generates diverse sentences, however topically similar to the input.
- DAE - latent space has empty regions

Deterministic AE	Variational AE
Input Sentence: a dog with its mouth open is running .	
a dog with its mouth is open running . a dog with its mouth is open running . a dog with its mouth is open running .	a dog with long hair is eating . a guy and the dogs are holding hands a dog with a toy at a rodeo .
Input Sentence: there are people sitting on the side of the road	
there are people sitting on the side of the road there are people sitting on the side of the road there are people sitting on the side of the road	the boy is walking down the street . there are people standing on the street outside the police are on the street corner .



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CVAEs [Sohn et al., 2015]

- Regular VAE - no control over the *class* of data being generated
- CVAEs - flexibility to synthesize data from the desired *class*

Minimize Reconstruction Loss + KL Divergence

$$J = \sum_{n=1}^N \left[-\mathbb{E}_{z^{(n)} \sim q} [\log p(x^{(n)} | z^{(n)}, c^{(n)})] + \lambda \cdot \text{KL}(q(z^{(n)} | x^{(n)}, c^{(n)}) \| p(z | c^{(n)})) \right]$$

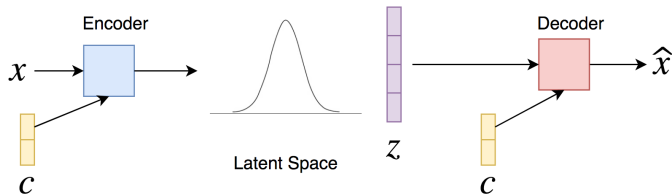


Figure: CVAE Model Architecture

Hypothesis for Outlier Detection

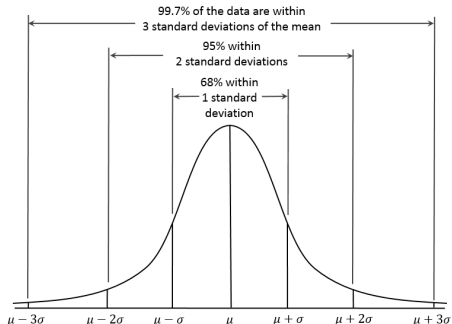


Figure: Univariate Normal distribution

- Data points further away from the mean are less probable
- More likely to be outliers
- For Apollo: Novel news detection
- Why CVAE: News articles conditioned on specific companies or sectors or even news history

Preliminary Experiments with MNIST

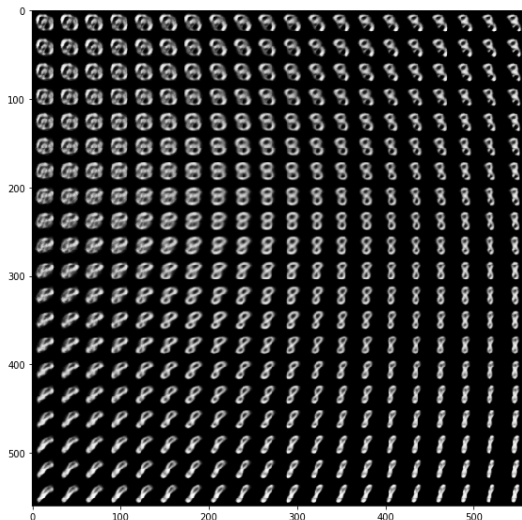


Figure: Samples from conditional prior of digit '8' - Blurry images when sampled away from the mean (centre)

Preliminary Experiments with MNIST

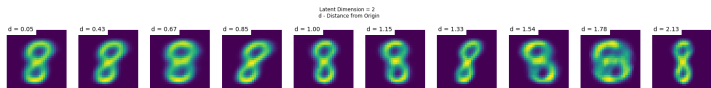


Figure: Sampling at different distances from the mean (origin)

Text Data

Yahoo Questions Dataset

- Conditioning Variable - Topic Label Embedding
- Subset of 100k questions

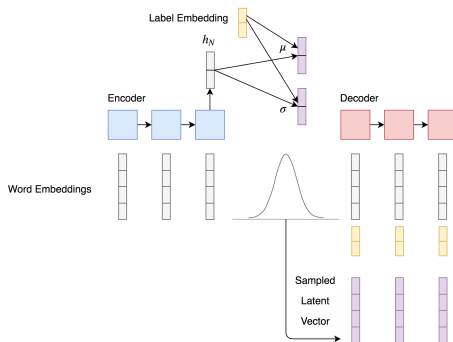


Figure: CVAE Model Architecture

Conditionally Generated Text

Samples drawn from the conditional posterior distribution of two topics:

Health	Sports
<i>how would i find my molar reaction ?</i> <i>what type of oily skin ?</i> <i>how does spinach go for the fat and vegetable ?</i> <i>how to control the swelling for this burning ?</i> <i>what is that mental disorder that i have one ?</i>	<i>can you find a alternative mountain bike ?</i> <i>why is the superbowl so amazing by brand ?</i> <i>whats your favorite swim team on each ?</i> <i>why is the boxing championships ?</i> <i>what is a club to be playing from the computer ?</i>

Sentences and Distances

- Created *fake* questions
- Topic: Health

Table: Success Cases

<i>how do you get rid of herpes ?</i> <i>do lawyers cause herpes ?</i>	28.8 42.1
<i>how soon can you know if you are pregnant ?</i> <i>how soon can you touch fresh paint ?</i>	43.2 53.4
<i>how can i grow my hair back ?</i> <i>are bald people good at doing business ?</i>	42.0 50.7

- Lower distances for shorter sentences
- Use of rare words results in higher distances

Table: Failure Cases

<i>how old were you before you were able to grow a good looking beard ?</i> <i>do you need a hammer to construct a good looking beard ?</i>	63.5 56.5
<i>how to relieve severe itchy skin ?</i> <i>how do police relieve severe criminals ?</i>	52.4 47.9
<i>should human genetic engineering be allowed ?</i> <i>should human build artificial intelligence ?</i>	47.1 38.4

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Conclusions and Future Work

• Summary

- VAEs are generative models from which it is possible to synthesize new data
- The usage of CVAEs for novelty/anomaly detection based on euclidean distance

• Issues

- For VAEs with textual data, the basis for clustering probably has to do more with syntax rather than semantics
- Gaussian latent space and euclidean distance may not be appropriate in high dimensions

• Next Steps

- Spherical VAEs based on von Mises Fisher Distribution - data is distributed on a unit hypersphere - cosine similarity as distance metric
[Xu and Durrett \[2018\]](#)

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